

Course : SWE 231: Software Requirement Engineering

Credit Hour : 3.0

Pre-requisite (if any) : SWE231

Course Synopsis : Software requirements engineering is the process of determining what is to be produced in a software system. It has the widely recognized goal of determining the needs for, and the intended external behavior, of a system design. This course will give participants a number of best practices, based upon the Software Engineering Body of Knowledge (SWEBOK) so that they can improve the quality of the requirements elicitation and development process in their organization. Characteristics of high-quality requirements specifications are completeness, conciseness, accuracy, modularization, prioritization, analysis, and verification. Ultimately, the techniques presented reduce project risk, improve product quality, and allow for effective control of requirements volatility – all measures for increasing the likelihood of a successful software project.

Mapping of Course Objectives, Learning Outcomes (LO), Teaching Learning (T-L) Activities and Assessments:

	Course objectives	Course learning outcomes	Teaching-learning activities	Assessment
1	Provide an overview of different Software Requirements types (functional vs non-functional (quality), constraints, business requirements, business rules, user and system requirements);	Have a clear understanding of processes, tools, and techniques used in requirements engineering;	Lectures	Quizzes
2	Describe different software requirements elicitation modalities;	Understand the concepts of software requirements elicitation, modeling, validation, and verification;	Demonstration, Group work	Quizzes
3	Provide approaches for requirements analysis and verification & validation;	Be able to model software requirements rigorously according to the latest requirements engineering	Demonstration, Group work, Assignment	Quizzes, Presentation

		standards;		
4	Provide approaches to manage requirements prioritization;	Be able to conduct a prioritization process for software requirements according to different approaches;	Lectures, Demonstration, Assignment	Quizzes, Lab Test
5	Provide approaches for software requirements effort estimation;	Be able to make a reasoned choice about the best approach for requirements modeling given the context of a project;	Lectures, Demonstration, Assignment	Quizzes, Presentation
6	Describe how to decompose system models: abstraction, & different system views for the definition of the software architecture from the requirements;	Understand the differences between different requirements modeling approaches (agile and non-agile); Be able to generate and maintain a software requirements specification document	Lectures, Demonstration, Group work, Assignment	Quizzes, Presentation

Mapping of Course LO and Generic Skills:

Learning Outcome (LO) of the course	Generic Skills* (Appendix-1)									
	1	2	3	4	5	6	7	8	9	10
1	√	√						√		√
2	√		√					√		√
3	√		√				√	√		√
4	√		√					√		√
5	√		√		√		√	√		√
6	√		√		√		√	√		√

Course Contents:

Lecture Contents	
Week-1-2	The Business Case for Requirements Analysis, Requirements Analysis through Software Life Cycles, Requirements Analysis based on the Nature of Software Development, Requirement Specification,
	Quizzes, Demonstration Group work
Week-3-4	Quality Assurance Methods, The Nature of Meetings, Understanding Requirements, System Planning Approaches, Requirements Validation and Testing,
	Class Test
Week-5-7	System Scope Models, Universal Modeling Language (UML), The Requirements Document, The Specifications Document, Software Tools Assisting the Development of Requirements and Specifications.
	Presentation, Demonstration Group work
Week-8-10	User Interface Design, Data Flow Modeling, Viewpoint Oriented Requirements Methods
Week-11-12	Non-Functional Requirements (Performance, Safety-Critical Systems)
	Class Test and Assignment (Team work)
Week-13-14	Introduction to Formal Methods, Formal Methods in Industrial Applications, Underpinnings of Formal Methods, Z and B for producing specifications.

Text Books:

1. Software Requirements Analysis and Specifications – Jag Sodh
2. Software Requirements Engineering–Richard H. Thayer, Merlin Dorfman, Sidney C. Bailin
3. Innovations for Requirement Analysis–Barbara Paech, Craig Martell